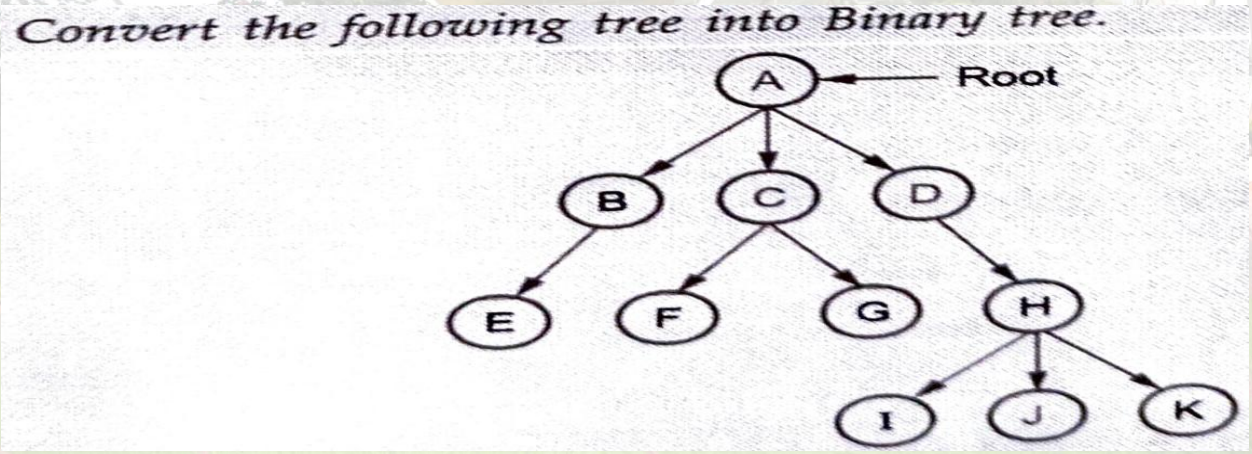
Mumbai Education Trust's

INSTITUTE OF ENGINEERING, NASHIK.

COMPUTER ENGINEERING DEPARTMENT

Subject : DS **ASSIGNMENT NO – 05** **Unit - IV**

1. Define **Basic Terminologies of Trees** with example and Explain **Binary tree with its Types(examples)**.
2. Define **Basic Terminologies of Graph** with example and Explain **Types of graph.**
3. Explain **Tree Traversal(all traversal) with it recursive algorithms and example**.
4. Write an algorithm for **non recursive post order traversal of binary tree.**
5. Describe and Write an **algorithm to delete a node from binary search tree**.
6. Write **pseudo code DFS & BFS traversal of binary search tree.**
7. Write an **algorithm for depth first traversal & Breadth First Traversal of a graph.**
8. With suitable example , Explain step for **conversion of a general tree into binary tree.**
   Note : Consider root as A



9. Explain the **Binary Search Tree in details with following example.**
   Contruct Binary Search Tree(BST) for the following :
   a) J, R, D, G, T, E, M, H, P, A, F, Q
   b) MAR, MAY, NOV. AUG, APR, JAN, DEC, JUL, FEB, JUN, OCT, SEPT

10. The **following numbers are inserted into an empty binary search tree in the given order :** G, C, B, A , D, E, F, I, H. Construct tree step by step. Represent the constructed tree using static memory allocation.

11.

a) From the given traversals construct the binary tree :

Inorder : EICFJBGDKHL **postorder :** IEJFCGKLHDB

Draw the binary tree and write preorder traversal.

b) From the given traversals construct the binary tree.

Pre-order : G, B, Q, A, C, K, F, P, D, E, R, H

In-order : Q, B, K, C, F, A, G, P, E, D, H, R

c)

Generate binary tree for the following pre-order and in-order traversals :

Pre-order : E A C K F H D B G

In-order : F A E K C D H G B

d) Construct a binary tree from its given preorder traversal PQSXYTRUV and postorder traversal XYSTQUVRP and apply inorder traversal to it.

e)

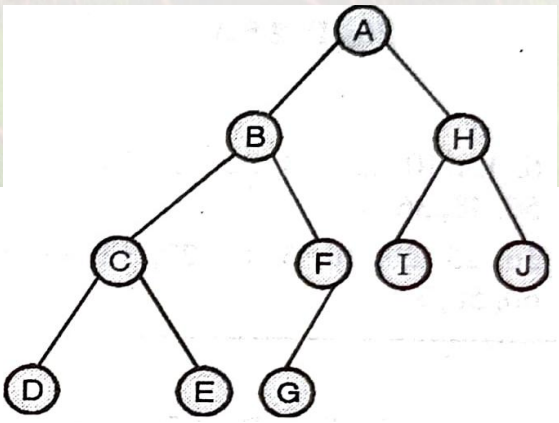
Construct a binary tree from given *two* traversals : [6]

Inorder Traversal—1 2 3 14 7 10 11 40 30

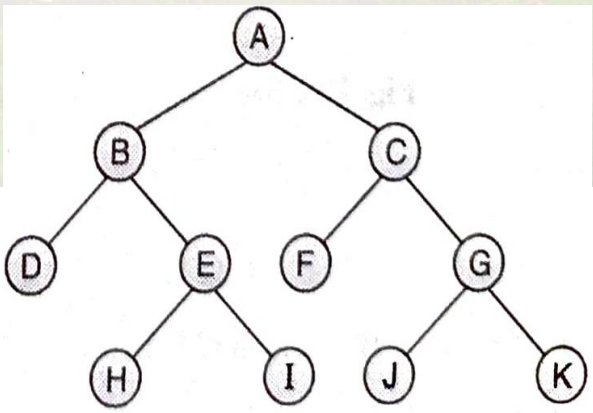
Postorder Traversal–1 3 2 7 10 40 30 11 14

12. **Solve following Tree traversal examples. Perform inorder , preorder and postorder traversal of binary tree.**

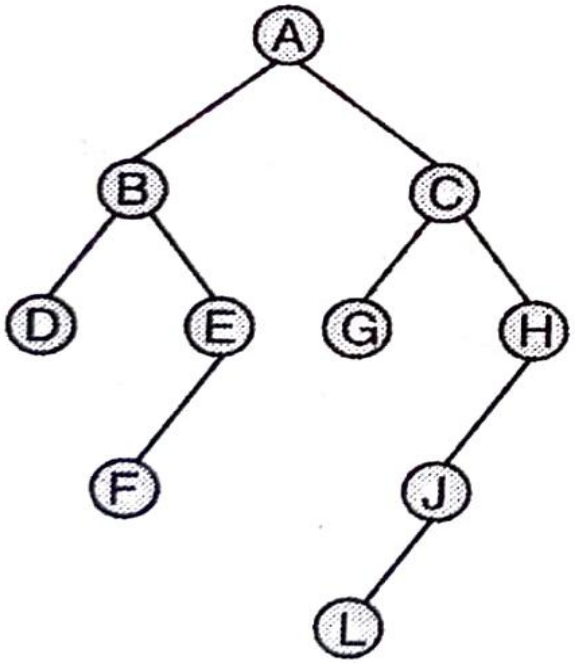
1.



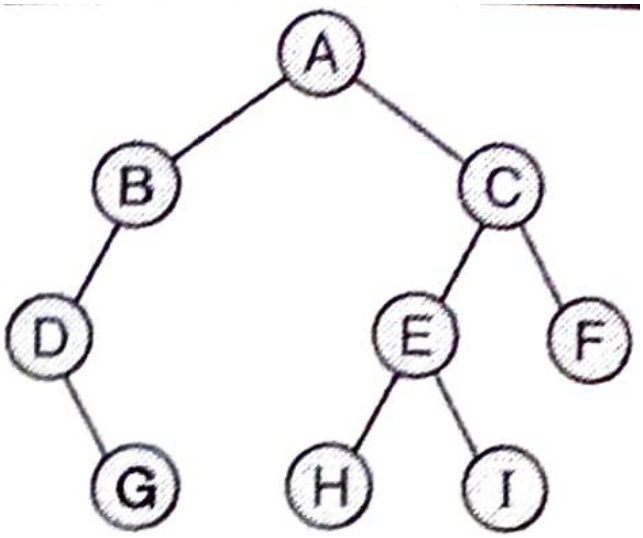
2.



3.



4.



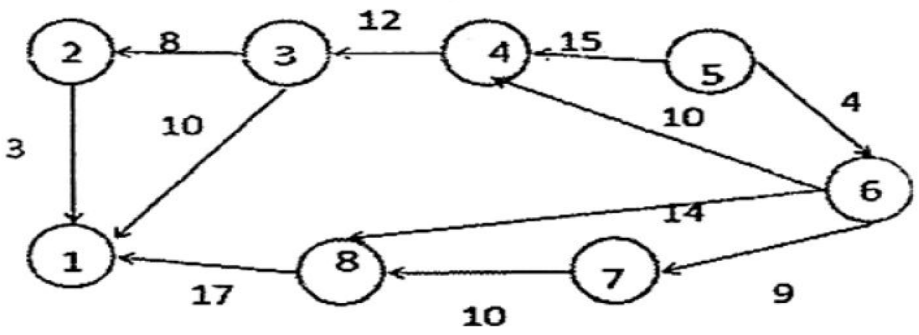
13. **Draw any directed graph with minimum 6 nodes and represent graph using adjacency matrix, adjacency list and inverse adjacency list.**

14.

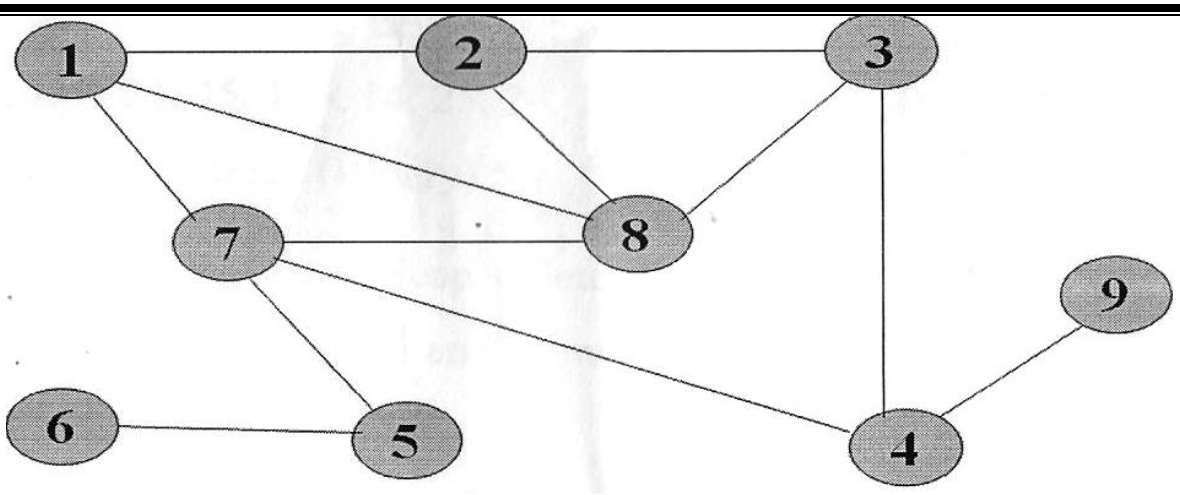
Explain the following terms of the below given graph :

(*i*) In degree of and out degree of each vertex.

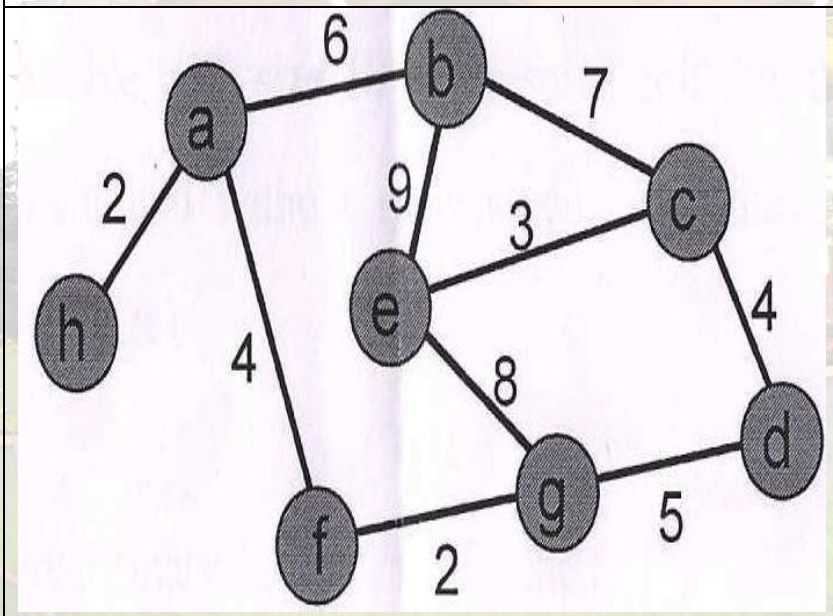
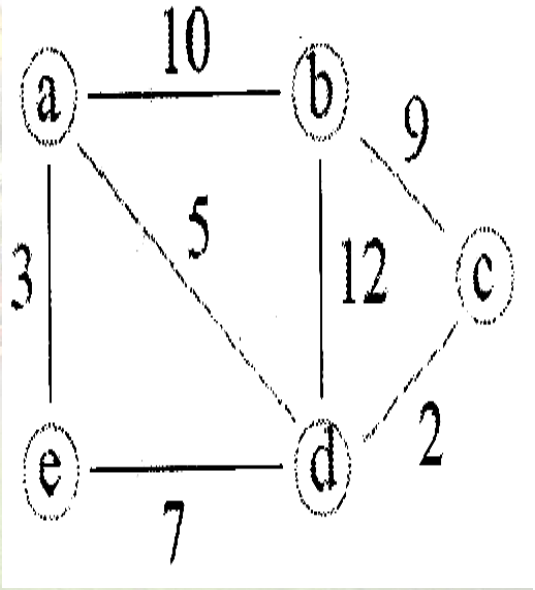
(*ii*) Adjancy list representation.



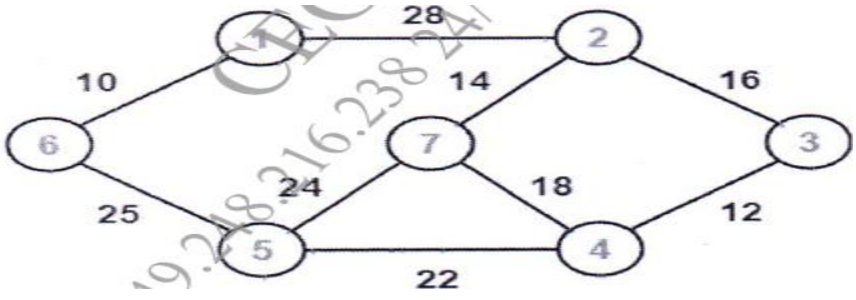
15. Define DFS and BFS for a graph. Show BFS and DFS for the following graph with starting vertex as 1.



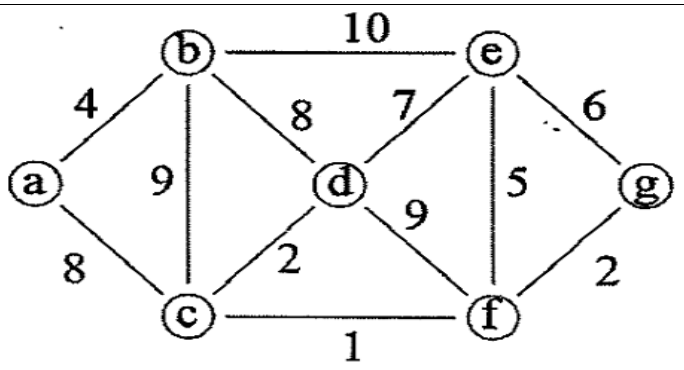
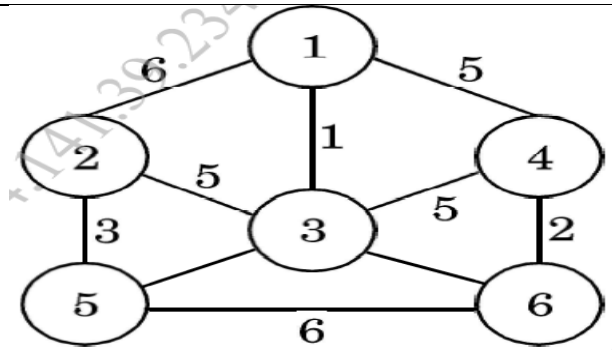
16. Write **Kruskal's algorithm for minimum spanning trees and explain with example. Find the MST for the graph given using Kruskals Algorithm and show all the Step**.

Example 1 :	**Example 2 :**
	

17. Construct the **minimum spanning tree (MST) for the given graph using Prim's Algorithm** staring from vertex 6.



18. Construct MST from the given data using Prims Algorithms.

Example 1:	**Example 2:**
	

19. Consider the graph represented by the following adjacency matrix :

And find minimum spanning tree of this graph using Prim's algorithm.

	1	2	3	4	5	6
1	0	3	1	6	0	0
2	3	0	5	0	3	0
3	1	5	0	5	6	4
4	6	0	5	0	0	2
5	0	3	6	0	0	6
6	0	0	2	2	6	0

********** **Best of Luck** **********

Study material provided by: Vishwajeet Londhe

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